JSO — resultados CEC

Benchmark MAREA · paper OE1 · 71 funciones CEC2014/2017/2022 · 51 corridas por funcion

Protocolo

Valores = **error al optimo**, f(x) - f*. Menor es mejor; 0 es el optimo global. Dominio [-100,100] en todas las funciones. Poblacion N=80, iteraciones T=MaxFES/N. Presupuestos 5.000 / 50.000 / 500.000 evaluaciones. Las 30 funciones de CEC2014 y las 29 de CEC2017 corren en D=10 y D=50; las 12 de CEC2022 solo llegan a D=20, asi que en la columna D50 corren en **D=20** (marcado en la tabla). Todos los resultados vienen del mismo entorno del Mac mini (python 3.11.9, numpy 2.3.3, opfunu 1.0.4).

Nota sobre este algoritmo

jSO con su propia poblacion (reduccion lineal), no N=80. Presupuesto (T+1)*N = 5040 / 50 080 / 500 080, igual que los clasicos.

Resumen — posicion entre los 10 algoritmos (rango por media)

dimension	presupuesto	rango medio de Friedman	posicion de 10	funciones donde es el mejor
D10	5k FES	2.62	2º	36 de 71
D10	50k FES	2.32	1º	33 de 71
D10	500k FES	2.50	1º	36 de 71
D50	5k FES	3.72	4º	6 de 71
D50	50k FES	2.19	1º	42 de 71
D50	500k FES	1.92	1º	39 de 71

JSO — estadisticas por funcion, D = 10

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	10	5.75e+02	3.44e+04	2.73e+03	5.18e+03	7.19e+03	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F22014	2014	10	4.02e+02	2.33e+04	4.31e+03	5.69e+03	4.66e+03	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F32014	2014	10	7.22e-02	1.06e+01	1.60e+00	2.46e+00	2.25e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F42014	2014	10	2.56e-01	3.48e+01	3.48e+01	3.05e+01	1.11e+01	0.00e+00	3.48e+01	3.48e+01	3.14e+01	1.04e+01	0.00e+00	3.48e+01	3.48e+01	3.14e+01	1.04e+01
F52014	2014	10	2.03e+01	2.08e+01	2.06e+01	2.06e+01	1.19e-01	1.98e-06	2.05e+01	2.03e+01	1.89e+01	4.44e+00	0.00e+00	2.00e+01	0.00e+00	8.23e+00	9.94e+00
F62014	2014	10	3.70e-01	5.36e+00	1.09e+00	1.28e+00	8.52e-01	0.00e+00	1.14e-13	0.00e+00	2.23e-15	1.59e-14	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F72014	2014	10	1.85e-01	8.74e-01	5.37e-01	5.45e-01	1.87e-01	0.00e+00	1.48e-02	0.00e+00	3.24e-03	4.79e-03	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F82014	2014	10	1.50e+01	4.02e+01	2.81e+01	2.75e+01	5.97e+00	0.00e+00	3.98e+00	0.00e+00	1.37e-01	6.30e-01	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F92014	2014	10	2.00e+01	4.96e+01	3.58e+01	3.45e+01	8.58e+00	2.84e-08	4.97e+00	2.00e+00	2.29e+00	8.95e-01	0.00e+00	2.98e+00	1.99e+00	1.72e+00	7.20e-01
F102014	2014	10	4.11e+02	1.49e+03	1.15e+03	1.15e+03	2.05e+02	3.56e-04	5.84e-01	1.44e-01	1.65e-01	9.66e-02	0.00e+00	9.09e-13	0.00e+00	3.74e-13	4.52e-13
F112014	2014	10	1.04e+03	1.91e+03	1.56e+03	1.53e+03	1.94e+02	4.37e-01	3.89e+02	3.19e+01	7.61e+01	8.95e+01	1.87e-01	1.29e+02	1.17e+01	1.66e+01	2.03e+01
F122014	2014	10	9.22e-01	2.34e+00	1.72e+00	1.79e+00	3.45e-01	1.72e-01	1.29e+00	4.73e-01	5.79e-01	3.32e-01	2.82e-03	6.32e-02	2.53e-02	2.67e-02	1.19e-02
F132014	2014	10	1.37e-01	4.27e-01	3.06e-01	2.93e-01	6.24e-02	5.47e-02	1.49e-01	9.40e-02	9.26e-02	2.18e-02	7.25e-03	7.80e-02	3.80e-02	3.78e-02	1.46e-02
F142014	2014	10	1.33e-01	4.53e-01	3.43e-01	3.33e-01	7.18e-02	2.96e-02	2.18e-01	9.42e-02	1.00e-01	3.78e-02	7.48e-03	4.63e-02	2.62e-02	2.71e-02	8.84e-03
F152014	2014	10	1.26e+00	4.58e+00	3.52e+00	3.51e+00	5.94e-01	2.09e-01	7.67e-01	5.06e-01	5.11e-01	1.25e-01	1.62e-01	4.26e-01	3.00e-01	3.00e-01	5.79e-02
F162014	2014	10	3.16e+00	3.93e+00	3.55e+00	3.55e+00	1.79e-01	5.42e-01	2.20e+00	1.57e+00	1.45e+00	4.10e-01	2.07e-01	1.20e+00	5.24e-01	5.42e-01	2.50e-01
F172014	2014	10	2.15e+01	1.20e+03	2.91e+02	3.52e+02	2.57e+02	0.00e+00	2.08e-01	4.55e-13	4.90e-02	8.92e-02	0.00e+00	2.08e-01	0.00e+00	4.49e-02	8.65e-02
F182014	2014	10	1.76e+01	2.67e+02	4.04e+01	5.08e+01	3.86e+01	8.69e-02	5.00e-01	4.64e-01	3.96e-01	1.29e-01	2.79e-03	5.00e-01	1.15e-01	1.79e-01	1.70e-01
F192014	2014	10	3.71e+00	1.57e+01	5.71e+00	6.04e+00	2.05e+00	1.92e+00	3.02e+00	2.46e+00	2.45e+00	2.89e-01	1.16e-01	1.17e+00	5.75e-01	5.83e-01	2.53e-01
F202014	2014	10	2.22e+01	4.03e+03	1.48e+02	3.83e+02	7.55e+02	1.17e-02	6.13e-01	4.71e-01	4.27e-01	1.70e-01	2.97e-02	5.41e-01	1.59e-01	2.04e-01	1.46e-01
F212014	2014	10	2.89e+00	1.02e+03	3.45e+01	9.24e+01	1.53e+02	1.42e-03	1.12e+00	4.92e-01	4.46e-01	2.75e-01	3.10e-06	8.12e-01	1.49e-03	1.40e-01	2.11e-01
F222014	2014	10	2.71e+01	2.50e+02	3.45e+01	4.38e+01	4.21e+01	6.71e-01	2.33e+02	2.12e+01	1.97e+01	3.17e+01	9.88e-02	2.06e+01	3.25e-01	5.82e+00	9.06e+00
F232014	2014	10	2.37e+02	2.42e+02	2.42e+02	2.42e+02	7.55e-01	2.37e+02	2.42e+02	2.42e+02	2.42e+02	1.62e+00	2.37e+02	2.42e+02	2.42e+02	2.42e+02	1.28e+00
F242014	2014	10	1.16e+02	1.51e+02	1.38e+02	1.37e+02	7.11e+00	1.00e+02	1.11e+02	1.07e+02	1.06e+02	2.60e+00	1.00e+02	1.08e+02	1.06e+02	1.05e+02	2.74e+00
F252014	2014	10	1.83e+02	2.00e+02	2.00e+02	2.00e+02	3.66e+00	1.61e+02	2.00e+02	2.00e+02	1.98e+02	7.87e+00	1.50e+02	2.00e+02	2.00e+02	1.98e+02	9.67e+00
F262014	2014	10	1.00e+02	1.00e+02	1.00e+02	1.00e+02	6.43e-02	1.00e+02	1.00e+02	1.00e+02	1.00e+02	2.05e-02	1.00e+02	1.00e+02	1.00e+02	1.00e+02	1.44e-02
F272014	2014	10	5.81e+00	4.01e+02	4.00e+02	2.76e+02	1.82e+02	1.75e+00	4.00e+02	4.00e+02	2.13e+02	2.00e+02	2.18e-01	4.00e+02	1.52e+00	1.89e+02	2.01e+02
F282014	2014	10	3.58e+02	4.88e+02	3.72e+02	3.88e+02	3.88e+01	3.57e+02	4.81e+02	3.69e+02	3.82e+02	3.45e+01	3.57e+02	4.81e+02	3.72e+02	3.79e+02	2.96e+01
F292014	2014	10	4.50e+06	7.49e+06	5.31e+06	5.18e+06	6.07e+05	4.50e+06	7.49e+06	4.89e+06	4.96e+06	5.40e+05	4.50e+06	6.86e+06	4.89e+06	4.99e+06	5.78e+05
F302014	2014	10	2.79e+03	5.97e+04	5.12e+03	1.86e+04	2.00e+04	2.31e+03	4.90e+04	2.31e+03	8.77e+03	1.62e+04	2.31e+03	4.90e+04	2.31e+03	5.07e+03	1.11e+04
F12017	2017	10	1.26e+03	5.03e+04	5.98e+03	8.41e+03	8.25e+03	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F22017	2017	10	3.20e-03	9.38e-01	3.60e-02	7.05e-02	1.37e-01	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F32017	2017	10	1.65e+00	8.86e+00	7.51e+00	6.59e+00	2.25e+00	0.00e+00	3.99e+00	3.99e+00	2.58e+00	1.92e+00	0.00e+00	3.99e+00	3.99e+00	2.42e+00	1.97e+00
F42017	2017	10	1.42e+01	6.43e+01	4.72e+01	4.67e+01	9.22e+00	2.80e-10	3.99e+00	1.99e+00	2.23e+00	9.68e-01	9.95e-01	2.98e+00	1.99e+00	1.85e+00	7.96e-01
F52017	2017	10	8.28e-06	1.52e-03	2.74e-04	3.97e-04	3.26e-04	1.14e-12	1.09e-08	3.05e-10	1.35e-09	2.25e-09	0.00e+00	5.85e-12	5.68e-14	4.40e-13	1.07e-12
F62017	2017	10	4.84e+01	8.20e+01	6.56e+01	6.56e+01	8.46e+00	1.12e+01	1.63e+01	1.23e+01	1.23e+01	9.08e-01	1.09e+01	1.30e+01	1.17e+01	1.19e+01	6.03e-01
F72017	2017	10	2.66e+01	7.65e+01	5.52e+01	5.32e+01	1.48e+01	2.00e+00	1.40e+01	8.00e+00	8.21e+00	2.19e+00	2.00e+00	8.00e+00	6.00e+00	5.53e+00	1.47e+00
F82017	2017	10	3.88e-06	1.27e-03	3.44e-05	9.05e-05	1.87e-04	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F92017	2017	10	7.82e+02	1.94e+03	1.49e+03	1.46e+03	2.04e+02	4.37e-01	3.07e+02	1.60e+01	3.67e+01	6.20e+01	1.25e-01	1.02e+01	3.66e+00	3.84e+00	2.68e+00
F102017	2017	10	1.64e+01	8.83e+02	4.73e+01	9.82e+01	1.48e+02	0.00e+00	9.95e-01	1.44e-11	1.95e-02	1.39e-01	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F112017	2017	10	6.40e+02	1.48e+04	3.43e+03	4.43e+03	3.25e+03	0.00e+00	2.08e-01	2.08e-01	1.31e-01	1.02e-01	0.00e+00	2.08e-01	2.08e-01	1.35e-01	1.00e-01
F122017	2017	10	3.48e+01	1.50e+03	1.93e+02	3.00e+02	3.03e+02	4.09e+00	7.47e+00	4.61e+00	4.71e+00	6.02e-01	0.00e+00	4.28e+00	4.09e+00	3.94e+00	8.07e-01
F132017	2017	10	3.05e+01	1.31e+02	5.15e+01	5.70e+01	2.01e+01	1.00e+00	2.45e+01	1.18e+01	1.27e+01	8.23e+00	0.00e+00	3.98e+00	6.59e-12	4.88e-01	7.28e-01
F142017	2017	10	6.37e+00	4.53e+02	5.66e+01	9.53e+01	9.46e+01	7.55e-04	5.00e-01	3.53e-01	2.98e-01	2.05e-01	3.74e-06	5.00e-01	1.58e-01	2.25e-01	2.15e-01
F152017	2017	10	1.27e+00	5.02e+02	5.21e+00	5.09e+01	1.22e+02	1.06e-03	7.28e-01	2.10e-01	2.50e-01	2.31e-01	6.97e-06	6.03e-01	1.12e-03	9.63e-02	1.36e-01
F162017	2017	10	3.58e+01	2.77e+02	5.41e+01	6.10e+01	3.33e+01	9.83e+00	3.01e+01	2.24e+01	2.30e+01	3.66e+00	2.02e-02	2.05e+01	2.61e+00	4.97e+00	5.92e+00
F172017	2017	10	2.37e+01	1.41e+02	4.11e+01	4.77e+01	2.24e+01	1.16e-03	2.06e+01	4.38e-01	1.50e+00	4.76e+00	2.38e-05	4.99e-01	1.89e-03	1.05e-01	1.70e-01
F182017	2017	10	6.57e+00	1.62e+01	1.03e+01	1.01e+01	2.13e+00	3.50e-01	2.01e+00	1.57e+00	1.37e+00	4.77e-01	8.43e-02	2.53e-01	1.31e-01	1.45e-01	4.43e-02
F192017	2017	10	4.16e+01	1.40e+02	7.10e+01	7.40e+01	2.29e+01	8.28e-01	2.78e+01	3.10e+00	4.23e+00	4.56e+00	2.88e-01	1.37e+00	7.55e-01	7.49e-01	2.47e-01

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	10	1.00e+02	1.03e+02	1.01e+02	1.01e+02	6.51e-01	1.00e+02	1.00e+02	1.00e+02	1.00e+02	2.27e-13	1.00e+02	1.00e+02	1.00e+02	1.00e+02	0.00e+00
F212017	2017	10	1.00e+02	1.04e+02	1.00e+02	1.00e+02	7.60e-01	1.00e+02	1.00e+02	1.00e+02	1.00e+02	2.28e-13	1.00e+02	1.00e+02	1.00e+02	1.00e+02	0.00e+00
F222017	2017	10	1.01e+02	2.04e+02	1.03e+02	1.12e+02	2.71e+01	1.00e+02	1.00e+02	1.00e+02	1.00e+02	0.00e+00	1.00e+02	1.00e+02	1.00e+02	1.00e+02	0.00e+00
F232017	2017	10	2.01e+02	2.08e+02	2.03e+02	2.03e+02	1.60e+00	2.00e+02	2.00e+02	2.00e+02	2.00e+02	0.00e+00	2.00e+02	2.00e+02	2.00e+02	2.00e+02	0.00e+00
F242017	2017	10	4.78e+02	5.07e+02	4.81e+02	4.81e+02	3.79e+00	4.76e+02	4.79e+02	4.79e+02	4.79e+02	1.24e+00	4.76e+02	4.79e+02	4.79e+02	4.79e+02	6.92e-01
F252017	2017	10	1.11e+00	4.68e+02	4.67e+02	4.48e+02	9.12e+01	0.00e+00	4.67e+02	4.67e+02	4.20e+02	1.40e+02	0.00e+00	4.68e+02	4.67e+02	4.48e+02	9.14e+01
F262017	2017	10	3.90e+02	4.00e+02	3.95e+02	3.94e+02	3.55e+00	3.90e+02	3.98e+02	3.91e+02	3.91e+02	2.45e+00	3.90e+02	3.97e+02	3.91e+02	3.91e+02	1.95e+00
F272017	2017	10	3.34e-01	6.91e+00	2.22e+00	2.39e+00	1.40e+00	0.00e+00	4.55e-13	0.00e+00	1.25e-13	2.05e-13	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F282017	2017	10	1.05e+03	1.46e+03	1.15e+03	1.16e+03	8.45e+01	9.79e+02	1.06e+03	1.05e+03	1.02e+03	3.02e+01	9.79e+02	1.04e+03	1.04e+03	1.02e+03	3.07e+01
F292017	2017	10	2.29e+03	1.67e+04	8.18e+03	8.16e+03	2.77e+03	1.38e+03	3.16e+03	1.38e+03	1.55e+03	5.34e+02	1.38e+03	3.16e+03	1.38e+03	1.48e+03	4.23e+02
F12022	2022	10	1.41e-03	1.78e-01	2.26e-02	3.24e-02	2.93e-02	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F22022	2022	10	2.80e-03	8.99e+00	6.63e+00	6.19e+00	3.13e+00	0.00e+00	8.92e+00	3.99e+00	5.49e+00	2.72e+00	0.00e+00	8.92e+00	3.99e+00	5.57e+00	2.61e+00
F32022	2022	10	2.82e-09	1.39e-06	9.99e-08	1.74e-07	2.20e-07	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F42022	2022	10	2.87e+01	8.00e+01	5.70e+01	5.60e+01	1.49e+01	6.00e+00	1.40e+01	1.00e+01	9.18e+00	2.23e+00	4.00e+00	1.40e+01	8.00e+00	7.41e+00	2.01e+00
F52022	2022	10	4.55e-07	3.71e-04	3.90e-05	5.39e-05	6.12e-05	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F62022	2022	10	4.56e+01	4.49e+03	4.57e+02	9.31e+02	1.08e+03	8.77e-02	5.00e-01	4.69e-01	4.08e-01	1.21e-01	2.81e-03	3.66e-01	4.17e-02	9.12e-02	1.02e-01
F72022	2022	10	3.08e+01	5.98e+01	4.69e+01	4.63e+01	7.18e+00	6.86e-01	2.88e+01	4.64e+00	1.08e+01	9.80e+00	4.52e-02	2.03e+01	2.77e-01	6.85e-01	2.81e+00
F82022	2022	10	1.77e+01	3.16e+01	2.78e+01	2.71e+01	2.99e+00	1.11e+00	6.11e+00	3.58e+00	3.48e+00	1.23e+00	1.35e-01	4.48e-01	2.63e-01	2.64e-01	7.13e-02
F92022	2022	10	3.51e-02	8.15e-01	1.51e-01	1.72e-01	1.14e-01	0.00e+00	4.55e-13	4.55e-13	2.50e-13	2.29e-13	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F102022	2022	10	2.21e+02	2.24e+02	2.22e+02	2.22e+02	8.33e-01	2.21e+02	2.21e+02	2.21e+02	2.21e+02	2.30e-13	2.21e+02	2.21e+02	2.21e+02	2.21e+02	2.87e-14
F112022	2022	10	6.87e-03	1.66e-01	3.71e-02	4.76e-02	3.18e-02	0.00e+00	4.55e-13	4.55e-13	2.50e-13	2.29e-13	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F122022	2022	10	1.65e+02	1.67e+02	1.66e+02	1.66e+02	7.28e-01	1.64e+02	1.67e+02	1.64e+02	1.65e+02	7.33e-01	1.64e+02	1.66e+02	1.64e+02	1.65e+02	6.11e-01

JSO — estadisticas por funcion, D = 50

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	50	8.83e+07	3.28e+08	1.71e+08	1.87e+08	5.79e+07	4.16e+05	1.70e+06	9.30e+05	9.49e+05	2.87e+05	1.70e-02	1.27e+02	1.69e+00	8.95e+00	2.15e+01
F22014	2014	50	5.66e+09	1.47e+10	9.26e+09	9.40e+09	1.72e+09	9.90e+02	9.24e+03	4.34e+03	4.85e+03	2.17e+03	0.00e+00	5.68e-14	2.84e-14	2.90e-14	6.94e-15
F32014	2014	50	6.55e+04	1.25e+05	8.67e+04	8.84e+04	1.29e+04	7.60e+00	1.13e+02	2.74e+01	3.55e+01	2.52e+01	0.00e+00	5.68e-14	5.68e-14	4.46e-14	2.36e-14
F42014	2014	50	9.11e+02	1.82e+03	1.39e+03	1.34e+03	2.53e+02	9.36e+01	1.03e+02	9.81e+01	9.77e+01	1.78e+00	2.65e-02	9.84e+01	9.81e+01	5.21e+01	4.93e+01
F52014	2014	50	2.12e+01	2.14e+01	2.13e+01	2.13e+01	3.98e-02	2.11e+01	2.13e+01	2.12e+01	2.12e+01	4.04e-02	2.05e+01	2.12e+01	2.11e+01	2.10e+01	1.86e-01
F62014	2014	50	5.21e+01	7.14e+01	6.16e+01	6.17e+01	4.58e+00	3.36e+00	1.44e+01	7.33e+00	7.09e+00	2.32e+00	1.49e-05	5.35e-01	1.09e-03	2.28e-02	1.03e-01
F72014	2014	50	4.95e+01	1.44e+02	1.04e+02	9.92e+01	2.37e+01	7.89e-04	2.99e-02	5.74e-03	8.54e-03	7.11e-03	0.00e+00	1.14e-13	0.00e+00	6.69e-15	2.70e-14
F82014	2014	50	3.98e+02	4.85e+02	4.48e+02	4.47e+02	2.08e+01	1.01e+02	2.36e+02	1.27e+02	1.34e+02	2.64e+01	1.28e-10	9.42e-08	4.01e-09	6.70e-09	1.35e-08
F92014	2014	50	3.82e+02	5.20e+02	4.60e+02	4.60e+02	2.64e+01	1.21e+02	2.80e+02	2.20e+02	2.16e+02	3.47e+01	1.06e+01	2.30e+01	1.78e+01	1.72e+01	2.89e+00
F102014	2014	50	1.23e+04	1.47e+04	1.40e+04	1.40e+04	4.16e+02	4.38e+03	6.30e+03	5.19e+03	5.25e+03	4.10e+02	4.15e+00	2.22e+01	9.48e+00	9.67e+00	3.78e+00
F112014	2014	50	1.39e+04	1.54e+04	1.48e+04	1.47e+04	3.54e+02	9.80e+03	1.37e+04	1.25e+04	1.23e+04	7.64e+02	2.48e+03	4.00e+03	3.30e+03	3.29e+03	3.26e+02
F122014	2014	50	3.51e+00	5.72e+00	4.77e+00	4.72e+00	5.25e-01	3.24e+00	4.61e+00	3.85e+00	3.92e+00	3.35e-01	1.91e-01	1.88e+00	2.86e-01	3.54e-01	3.11e-01
F132014	2014	50	6.04e-01	2.40e+00	1.04e+00	1.15e+00	4.73e-01	3.27e-01	5.76e-01	4.59e-01	4.52e-01	6.09e-02	1.07e-01	2.91e-01	1.98e-01	1.98e-01	3.13e-02
F142014	2014	50	9.69e+00	3.85e+01	2.49e+01	2.41e+01	7.06e+00	3.52e-01	7.49e-01	4.47e-01	4.51e-01	5.53e-02	1.57e-01	3.67e-01	2.90e-01	2.79e-01	4.92e-02
F152014	2014	50	5.92e+02	6.41e+03	2.22e+03	2.49e+03	1.48e+03	2.00e+01	2.72e+01	2.46e+01	2.43e+01	1.80e+00	4.72e+00	6.26e+00	5.52e+00	5.45e+00	3.59e-01
F162014	2014	50	2.27e+01	2.36e+01	2.34e+01	2.33e+01	1.77e-01	2.16e+01	2.30e+01	2.25e+01	2.24e+01	3.33e-01	1.45e+01	2.00e+01	1.69e+01	1.69e+01	7.75e-01
F172014	2014	50	3.78e+06	2.91e+07	1.20e+07	1.23e+07	5.96e+06	1.59e+04	6.61e+04	3.58e+04	3.60e+04	1.30e+04	2.48e+01	9.38e+02	6.95e+01	1.26e+02	1.61e+02
F182014	2014	50	2.02e+07	1.59e+08	8.00e+07	8.53e+07	3.38e+07	3.85e+04	8.60e+04	6.50e+04	6.56e+04	1.07e+04	3.84e+01	4.91e+02	1.61e+02	2.04e+02	1.23e+02
F192014	2014	50	1.12e+05	3.01e+06	8.18e+05	9.12e+05	6.12e+05	3.85e+01	4.88e+01	4.22e+01	4.25e+01	2.42e+00	2.61e+01	3.15e+01	2.92e+01	2.91e+01	1.06e+00
F202014	2014	50	4.68e+06	4.39e+09	8.52e+08	1.25e+09	1.13e+09	1.87e+04	6.34e+04	4.03e+04	4.00e+04	1.00e+04	4.16e+00	2.85e+01	9.64e+00	9.90e+00	3.47e+00
F212014	2014	50	5.13e+06	5.12e+07	1.63e+07	1.83e+07	9.14e+06	6.17e+03	4.01e+04	1.86e+04	1.96e+04	7.92e+03	4.71e+00	4.61e+02	1.09e+01	6.40e+01	1.09e+02
F222014	2014	50	7.21e+03	1.60e+08	9.13e+05	7.71e+06	2.34e+07	5.76e+01	4.31e+02	1.60e+02	2.06e+02	1.14e+02	2.35e+01	2.44e+02	2.66e+01	4.31e+01	5.89e+01
F232014	2014	50	2.43e+02	2.60e+02	2.50e+02	2.50e+02	3.28e+00	2.35e+02	2.35e+02	2.35e+02	2.35e+02	1.18e-05	2.35e+02	2.35e+02	2.35e+02	2.35e+02	2.87e-14
F242014	2014	50	3.23e+02	3.43e+02	3.32e+02	3.32e+02	4.95e+00	2.85e+02	2.95e+02	2.92e+02	2.92e+02	1.71e+00	2.75e+02	2.90e+02	2.86e+02	2.85e+02	4.05e+00
F252014	2014	50	2.38e+02	2.62e+02	2.47e+02	2.47e+02	4.60e+00	2.06e+02	2.08e+02	2.07e+02	2.07e+02	6.24e-01	2.05e+02	2.06e+02	2.05e+02	2.05e+02	2.18e-01
F262014	2014	50	1.03e+02	2.06e+02	1.06e+02	1.40e+02	4.70e+01	1.00e+02	2.00e+02	1.00e+02	1.02e+02	1.40e+01	1.00e+02	1.00e+02	1.00e+02	1.00e+02	3.16e-02
F272014	2014	50	1.37e+03	1.77e+03	1.60e+03	1.60e+03	7.69e+01	3.44e+02	5.01e+02	4.22e+02	4.21e+02	3.33e+01	3.00e+02	3.26e+02	3.20e+02	3.14e+02	1.08e+01
F282014	2014	50	2.17e+03	6.48e+03	3.48e+03	3.52e+03	6.56e+02	1.09e+03	1.37e+03	1.17e+03	1.18e+03	4.98e+01	1.03e+03	1.14e+03	1.09e+03	1.09e+03	3.04e+01
F292014	2014	50	1.08e+08	4.65e+08	2.33e+08	2.45e+08	8.81e+07	8.90e+03	2.72e+04	1.34e+04	1.39e+04	3.06e+03	4.20e+03	6.38e+03	4.65e+03	4.72e+03	3.87e+02
F302014	2014	50	5.00e+08	6.66e+09	2.48e+09	2.75e+09	1.55e+09	7.62e+04	3.54e+06	5.10e+05	6.17e+05	5.40e+05	1.43e+04	3.92e+04	1.74e+04	1.81e+04	3.60e+03
F12017	2017	50	4.36e+09	1.31e+10	8.87e+09	8.68e+09	1.95e+09	3.11e+02	9.75e+03	3.13e+03	3.45e+03	2.24e+03	1.42e-14	2.84e-14	1.42e-14	1.56e-14	4.27e-15
F22017	2017	50	3.70e+04	8.51e+04	5.64e+04	5.61e+04	1.05e+04	1.71e-02	3.38e-01	7.18e-02	8.40e-02	5.77e-02	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F32017	2017	50	6.82e+02	1.97e+03	1.13e+03	1.17e+03	2.91e+02	5.69e+01	1.62e+02	1.52e+02	1.41e+02	2.55e+01	5.61e+01	1.56e+02	5.61e+01	9.79e+01	4.88e+01
F42017	2017	50	7.58e+03	1.66e+04	1.17e+04	1.15e+04	2.46e+03	1.42e+02	3.05e+02	2.45e+02	2.40e+02	3.57e+01	5.07e+00	1.87e+01	1.31e+01	1.24e+01	2.92e+00
F52017	2017	50	1.26e-02	3.46e-02	2.27e-02	2.27e-02	5.53e-03	5.17e-08	1.68e-05	4.53e-07	1.86e-06	3.46e-06	0.00e+00	8.57e-10	8.34e-11	1.64e-10	2.01e-10
F62017	2017	50	2.01e+05	5.31e+05	3.75e+05	3.73e+05	6.91e+04	2.14e+02	4.11e+02	3.51e+02	3.41e+02	4.74e+01	5.64e+01	6.93e+01	6.31e+01	6.32e+01	2.83e+00
F72017	2017	50	7.69e+02	1.05e+03	9.48e+02	9.29e+02	6.65e+01	2.97e+02	6.02e+02	5.23e+02	5.20e+02	5.98e+01	4.90e+01	9.41e+01	6.63e+01	6.76e+01	9.07e+00
F82017	2017	50	7.14e+00	1.56e+01	1.10e+01	1.11e+01	2.10e+00	4.01e-06	8.95e-02	3.22e-05	1.80e-03	1.25e-02	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F92017	2017	50	1.38e+04	1.56e+04	1.51e+04	1.50e+04	3.79e+02	9.43e+03	1.44e+04	1.26e+04	1.26e+04	9.31e+02	2.00e+03	4.04e+03	3.42e+03	3.45e+03	4.14e+02
F102017	2017	50	5.58e+05	8.49e+06	2.40e+06	2.68e+06	1.57e+06	5.26e+02	7.29e+03	1.78e+03	2.25e+03	1.57e+03	1.89e-01	2.37e+01	1.62e+01	1.18e+01	7.84e+00
F112017	2017	50	1.79e+08	5.94e+08	3.13e+08	3.42e+08	9.82e+07	3.40e+04	4.23e+05	1.03e+05	1.17e+05	7.20e+04	2.33e+01	1.61e+03	3.97e+02	4.92e+02	3.61e+02
F122017	2017	50	1.62e+08	8.47e+08	3.78e+08	4.18e+08	1.74e+08	1.53e+04	8.00e+04	4.94e+04	5.01e+04	1.36e+04	8.05e+00	2.40e+02	3.52e+01	4.51e+01	3.74e+01
F132017	2017	50	4.34e+05	2.37e+06	1.06e+06	1.11e+06	4.27e+05	1.05e+03	1.06e+04	4.79e+03	4.94e+03	2.26e+03	2.01e+01	3.17e+01	2.52e+01	2.51e+01	2.22e+00
F142017	2017	50	2.25e+07	2.04e+08	5.54e+07	6.25e+07	3.44e+07	5.71e+03	3.00e+04	1.50e+04	1.59e+04	4.97e+03	1.02e+01	2.57e+01	1.48e+01	1.58e+01	3.51e+00
F152017	2017	50	9.74e+04	2.07e+06	3.64e+05	5.30e+05	4.53e+05	1.76e+01	1.18e+03	2.35e+02	2.09e+02	2.32e+02	4.96e+00	2.33e+02	1.45e+01	3.87e+01	7.05e+01
F162017	2017	50	4.84e+04	3.77e+07	1.77e+05	1.31e+06	5.37e+06	2.03e+02	1.58e+03	4.42e+02	5.35e+02	3.39e+02	2.32e+01	3.67e+01	2.65e+01	2.70e+01	2.18e+00
F172017	2017	50	8.26e+05	8.54e+06	2.78e+06	3.26e+06	1.73e+06	3.44e+03	2.88e+04	1.61e+04	1.58e+04	5.93e+03	2.05e+01	2.35e+01	2.09e+01	2.13e+01	7.75e-01
F182017	2017	50	4.41e+09	2.46e+10	9.42e+09	1.10e+10	5.46e+09	1.32e+02	2.33e+03	4.92e+02	6.80e+02	4.67e+02	9.56e+00	2.22e+01	1.39e+01	1.46e+01	3.32e+00
F192017	2017	50	7.47e+02	1.77e+03	1.33e+03	1.28e+03	2.54e+02	3.24e+01	2.70e+02	5.07e+01	7.77e+01	7.60e+01	2.19e+01	2.41e+02	2.37e+01	4.05e+01	5.89e+01

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	50	2.77e+03	1.35e+04	8.52e+03	8.33e+03	2.25e+03	1.01e+02	5.06e+02	4.42e+02	3.85e+02	1.37e+02	1.00e+02	2.17e+02	2.11e+02	1.97e+02	3.91e+01
F212017	2017	50	3.93e+02	5.51e+02	4.51e+02	4.56e+02	3.79e+01	1.91e+02	2.23e+02	2.03e+02	2.03e+02	6.56e+00	1.00e+02	2.01e+02	1.00e+02	1.21e+02	3.98e+01
F222017	2017	50	1.31e+04	2.51e+04	1.73e+04	1.78e+04	2.98e+03	2.01e+02	2.08e+02	2.03e+02	2.03e+02	1.37e+00	2.00e+02	3.09e+02	2.00e+02	2.02e+02	1.53e+01
F232017	2017	50	6.93e+03	1.26e+04	9.41e+03	9.52e+03	1.29e+03	2.01e+02	2.04e+02	2.02e+02	2.02e+02	6.87e-01	2.00e+02	2.00e+02	2.00e+02	2.00e+02	0.00e+00
F242017	2017	50	1.82e+03	3.52e+03	2.78e+03	2.73e+03	4.00e+02	7.03e+02	7.83e+02	7.33e+02	7.34e+02	1.88e+01	7.06e+02	7.44e+02	7.25e+02	7.22e+02	1.05e+01
F252017	2017	50	2.18e+03	2.97e+03	2.59e+03	2.57e+03	2.10e+02	1.14e+03	1.62e+03	1.22e+03	1.27e+03	1.40e+02	1.18e+03	1.62e+03	1.20e+03	1.34e+03	1.66e+02
F262017	2017	50	7.74e+02	1.19e+03	9.91e+02	9.94e+02	9.12e+01	4.84e+02	5.18e+02	5.01e+02	5.00e+02	7.60e+00	4.68e+02	5.13e+02	4.79e+02	4.80e+02	8.41e+00
F272017	2017	50	5.43e+02	6.91e+02	6.10e+02	6.17e+02	3.49e+01	1.62e+00	3.45e+01	1.74e+01	1.67e+01	8.35e+00	4.55e-13	9.09e-13	4.55e-13	4.99e-13	1.37e-13
F282017	2017	50	2.39e+08	5.20e+09	1.13e+09	1.49e+09	9.98e+08	5.94e+03	3.37e+04	1.02e+04	1.16e+04	5.27e+03	2.97e+03	6.27e+03	3.64e+03	3.88e+03	7.42e+02
F292017	2017	50	1.32e+09	1.04e+10	4.75e+09	5.01e+09	2.14e+09	6.61e+04	2.56e+05	1.18e+05	1.23e+05	3.26e+04	6.06e+03	2.36e+04	8.55e+03	9.54e+03	4.23e+03
F12022	2022	20	7.72e+01	9.96e+02	2.49e+02	2.85e+02	1.54e+02	0.00e+00	5.68e-14	5.68e-14	3.57e-14	2.78e-14	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F22022	2022	20	5.03e+01	7.62e+01	5.79e+01	5.95e+01	6.72e+00	4.49e+01	4.91e+01	4.91e+01	4.74e+01	2.07e+00	4.49e+01	4.91e+01	4.49e+01	4.65e+01	2.07e+00
F32022	2022	20	1.85e-04	2.44e-03	8.11e-04	8.55e-04	4.64e-04	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F42022	2022	20	1.58e+02	2.83e+02	2.42e+02	2.39e+02	2.86e+01	1.81e+01	9.73e+01	5.29e+01	5.39e+01	2.16e+01	1.00e+01	3.20e+01	2.00e+01	1.98e+01	4.25e+00
F52022	2022	20	5.09e-02	3.26e-01	1.40e-01	1.40e-01	5.51e-02	0.00e+00	1.14e-13	0.00e+00	2.23e-14	4.56e-14	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F62022	2022	20	3.12e+04	5.96e+05	1.91e+05	2.27e+05	1.29e+05	1.50e+00	4.86e+02	4.30e+01	8.39e+01	1.12e+02	4.40e-01	5.00e-01	4.94e-01	4.89e-01	1.27e-02
F72022	2022	20	5.86e+01	2.32e+02	9.72e+01	1.00e+02	2.73e+01	2.64e+01	3.57e+01	3.01e+01	2.99e+01	1.99e+00	7.28e+00	2.17e+01	2.10e+01	2.04e+01	2.22e+00
F82022	2022	20	4.86e+01	1.15e+03	2.50e+02	3.53e+02	2.83e+02	8.01e+00	2.78e+01	2.66e+01	2.44e+01	5.54e+00	8.80e-01	2.43e+01	3.07e+00	4.98e+00	6.13e+00
F92022	2022	20	3.42e+02	3.76e+02	3.56e+02	3.57e+02	7.94e+00	3.36e+02	3.36e+02	3.36e+02	3.36e+02	6.02e-10	3.36e+02	3.36e+02	3.36e+02	3.36e+02	1.78e-13
F102022	2022	20	4.11e+02	6.26e+02	4.20e+02	4.26e+02	3.16e+01	3.97e+02	4.02e+02	3.99e+02	3.98e+02	1.72e+00	3.97e+02	3.97e+02	3.97e+02	3.97e+02	1.43e-01
F112022	2022	20	2.10e+00	1.45e+01	7.40e+00	7.88e+00	2.96e+00	3.64e-12	3.22e-10	2.09e-11	3.25e-11	4.64e-11	0.00e+00	4.55e-13	4.55e-13	4.19e-13	1.23e-13
F122022	2022	20	2.46e+02	2.87e+02	2.59e+02	2.61e+02	7.83e+00	2.35e+02	2.48e+02	2.38e+02	2.39e+02	2.20e+00	2.35e+02	2.48e+02	2.38e+02	2.39e+02	2.58e+00